Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Key idea: For a device with no other loads connected, the power it draws equals the product of the battery âĢĻs voltage and the current through it, P = V ÃĹ I, so I = P/V. The device âĢĻs resistance is then R = V/I.

- Given voltage V = 12 V and power P = 40 W:

I = 40 W / 12 V = 3.33 A (approximately).

- Then the required resistance is R = V/I = 12 V / 3.33 A âīĪ 3.60 Î©.

Therefore, the correct choice is E: 3.33 amps, 3.6 ohms.

</think>

<answer>

E. 3.33 amps, 3.6 ohms

</answer>